General Purpose Input Output

Introduction to ARM Cortex-M Microprocessors — Chapter 8

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Why This Chapter Matters

Chapter 4 blinked an LED in C. Chapters 5–7 showed what that program became: load, mask, store, branch. This chapter brings both views together. The same `MODER`/`ODR` pattern is the basis of every peripheral in this book. A GPIO pin is the simplest link between software and the physical world.

After this chapter, you can:

1. Describe the parts of a GPIO pin.
2. Choose push-pull or open-drain for a circuit.
3. Configure mode, type, speed and pulls.
4. Explain why the peripheral clock comes first.
5. Use `BSRR` for atomic pin changes.
6. Read a switch reliably, with debouncing.
7. Drive a display from a lookup table.

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By the end of this chapter you will know why these two lines are the safest way to switch a pin, and what each bit does:

```
GPIOA->BSRR = (1U << 5);          // PA5 high: LED on
GPIOA->BSRR = (1U << (5 + 16));   // PA5 low:  LED off
```

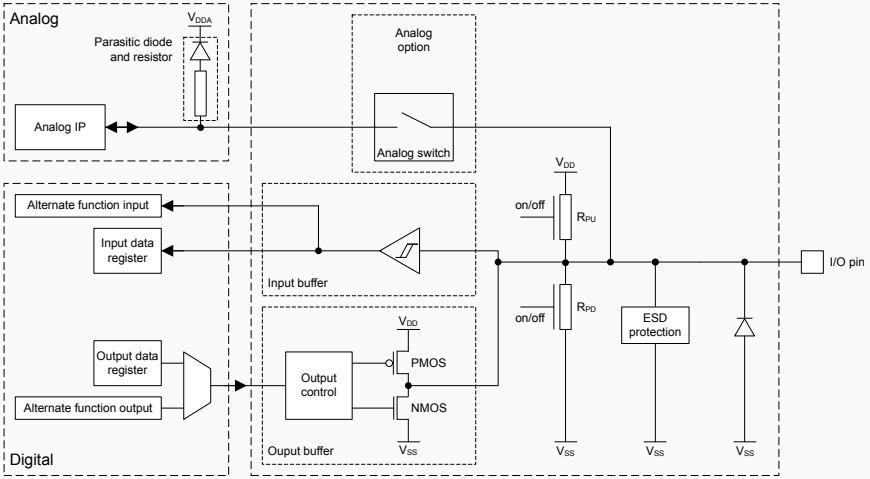
Questions this code raises, and where this chapter answers them:

Question raised	Answered in
What is BSRR?	The GPIO registers
Why not write ODR?	BSRR: atomic set and reset
What makes the pin able to drive?	Inside a GPIO pin
Why does a button need extra care?	Experiments: pull-ups and debouncing

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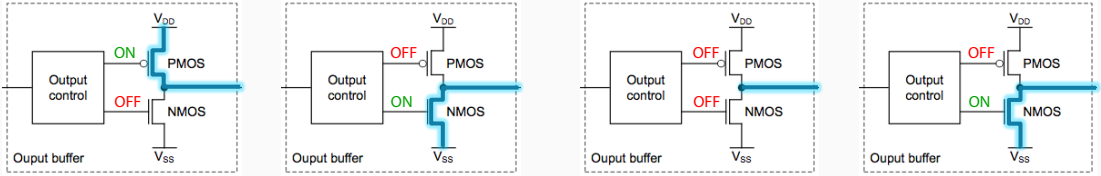
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Inside a GPIO Pin



One pin has two separate paths: an input path through a Schmitt trigger to `IDR`, and an output path from `ODR` through the driver transistors. Pull resistors sit in between.

Push-Pull and Open-Drain



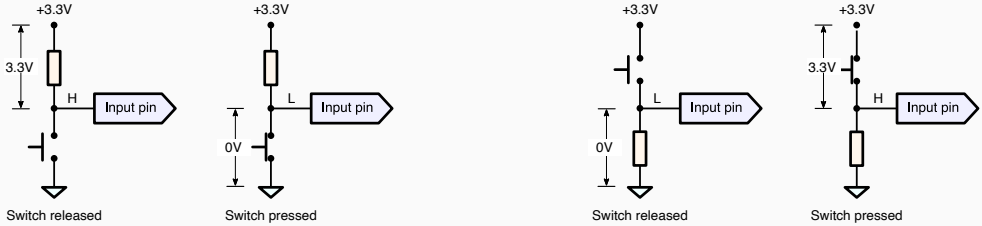
Push-pull — drives both high and low

Open-drain — pulls low, or releases

When you need open-drain

Push-pull is the default. It is what you want for an LED. But if two devices share one wire, one driving high while the other drives low causes a short circuit. Open-drain lets many devices share one line safely. This is exactly how I^2C works (Chapter 13).

Pull-Up and Pull-Down

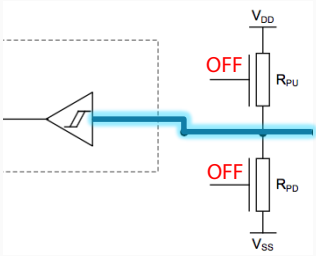


An unconnected input *floats*

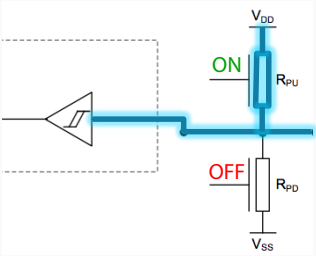
It picks up noise and reads a random value. A pull resistor gives it a fixed level when nothing else is driving it.

A button wired to ground needs a pull-up, so pressed reads 0. This is called active-low. PUPDR can turn on a pull-up inside the chip, so you do not need an extra part.

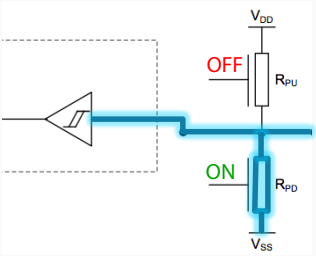
PUPDR — The Internal Resistors



Floating



Pulled up



Pulled down

Two bits per pin: `00` floating, `01` pull-up, `10` pull-down. An internal resistor means one fewer part on the board. This is Chapter 1's cost argument again.

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The Register Set

Register	Bits/pin	Purpose
`MODER`	2	Mode: input, output, alternate function, analog
`OTYPER`	1	Output type: push-pull or open-drain
`OSPEEDR`	2	Output slew rate
`PUPDR`	2	Pull-up, pull-down, or neither
`IDR`	1	Input data — read the pin
`ODR`	1	Output data — drive the pin
`BSRR`	2	Atomic bit set / reset
`AFR[2]`	4	Which alternate function (Chapter 10)

Mask width and bit position

1-bit field: mask `1`, shifted by n.
2-bit field: mask `3`, shifted by $2n$.
4-bit field: mask `0xF`, shifted by $4n$.

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MODER — Selecting the Mode

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MODER15[1:0]		MODER14[1:0]		MODER13[1:0]		MODER12[1:0]		MODER11[1:0]		MODER10[1:0]		MODER9[1:0]		MODER8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODER7[1:0]		MODER6[1:0]		MODER5[1:0]		MODER4[1:0]		MODER3[1:0]		MODER2[1:0]		MODER1[1:0]		MODER0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits	Mode
00	Input (reset state)
01	General-purpose output
10	Alternate function
11	Analog

```
GPIOA->MODER &= ~(3U « (5*2));  // clear
GPIOA->MODER |=  (1U « (5*2));  // set 01
```

Clear first, then set. Without the clear step, you cannot change a `11` into a `01`.

OTYPER — Output Type

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OT15	OT14	OT13	OT12	OT11	OT10	OT9	OT8	OT7	OT6	OT5	OT4	OT3	OT2	OT1	OT0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit	Output type
0	Push-pull (reset state)
1	Open-drain

```
GPIOA->OTYPER &= ~(1U « 5);  // push-pull
```

Unlike MODER, each pin owns only one bit: OT*y* for pin *y*. There are only two output types to choose from.

BSRR — Atomic Set and Reset

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
BR15	BR14	BR13	BR12	BR11	BR10	BR9	BR8	BR7	BR6	BR5	BR4	BR3	BR2	BR1	BR0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BS15	BS14	BS13	BS12	BS11	BS10	BS9	BS8	BS7	BS6	BS5	BS4	BS3	BS2	BS1	BS0
w	w	w	w	w	w	w	w	w	w	w	w	w	w	w	w

`ODR |= x` is a read-modify-write: three separate steps. If an interrupt changes another pin in between, that change is lost.

```
GPIOA->BSRR = (1U << 5);       // PA5 on
GPIOA->BSRR = (1U << (5+16)); // PA5 off
```

One write, no read. Note the = instead of `|=`. Using `|=` here would be a bug.

Sample I/O Addresses on the STM32F4 (1)

Every peripheral is reached the same way: a base address plus a fixed offset. This is the memory-map pattern from Chapter 2.

GPIO Port A		Clock Control (RCC)		Timer 2	
GPIOA_MODER	0x40020000	RCC_CR	0x40023800	TIM2_CR1	0x40000000
GPIOA_OTYPER	0x40020004	RCC_PLLCFGR	0x40023804	TIM2_CR2	0x40000004
GPIOA_OSPEEDR	0x40020008	RCC_CFGR	0x40023808	TIM2_DIER	0x4000000C
GPIOA_PUPDR	0x4002000C	RCC_AHB1ENR	0x40023830	TIM2_SR	0x40000010
GPIOA_IDR	0x40020010	RCC_APB1ENR	0x40023840	TIM2_EGR	0x40000014
GPIOA_ODR	0x40020014	RCC_APB2ENR	0x40023844	TIM2_CNT	0x40000024
GPIOA_BSRR	0x40020018			TIM2_PSC	0x40000028
GPIOA_AFRL	0x40020020			TIM2_ARR	0x4000002C

GPIO port registers sit `0x400` bytes apart. The `RCC` registers gate the clocks of all the others.

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Sample I/O Addresses on the STM32F4 (2)

ADC1		USART1		NVIC	
ADC_SR	0x40012000	USART1_SR	0x40011000	NVIC_ISER0	0xE000E100
ADC_CR1	0x40012004	USART1_DR	0x40011004	NVIC_ISER1	0xE000E104
ADC_CR2	0x40012008	USART1_BRR	0x40011008	NVIC_ICER0	0xE000E180
ADC_DR	0x4001204C	USART1_CR1	0x4001100C	NVIC_ICER1	0xE000E184

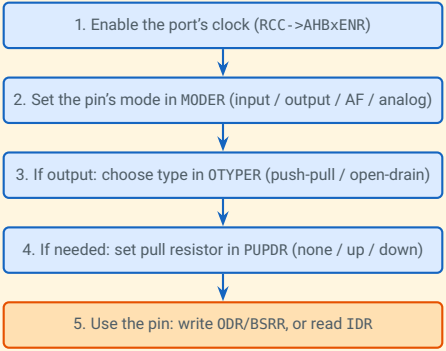
Read the leading digits as the memory map taught you: `0x40...` is a peripheral register. The NVIC sits at `0xE000...`, in the Private Peripheral region reserved for the core itself. GPIO, timer, ADC and USART registers share the `0x40` prefix because they are all ordinary memory-mapped peripherals.

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Enabling a GPIO Pin: The Conceptual Steps



Order matters

Step 1 must come first: a peripheral with no clock ignores every write. Steps 2–4 configure the pin *before* it is ever touched in the main loop — step 5 only ever reads or writes IDR/ODR/BSRR once setup is done.

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Peripheral Clock Gating

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	OTGHS ULPIEN	OTGHS EN	Res.	Res.	Res.	Res.	Res.	Res.	DMA2 EN	DMA1 EN	Res.	Res.	BKP SRAMEN	Res.	Res.
	rw	rw							rw	rw			rw		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	CRC EN	Res.	Res.	Res.	Res.	GPIOH EN	GPIOG EN	GPIOF EN	GPIOE EN	GPIOD EN	GPIOC EN	GPIOB EN	GPIOA EN
			rw					rw	rw	rw	rw	rw	rw	rw	rw

The classic first bug

Every peripheral starts with its clock disabled, to save power. A peripheral with no clock does not respond to writes: the writes are simply lost. No fault, no error, nothing happens.

```
RCC->AHB1ENR |= (1U « 0);  // GPIOA
```

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With ODR

```c
#include "stm32f4xx.h"
void delay_ms(uint32_t ms) {
    for (uint32_t i=0;i<ms*4000;i++) __NOP();
}
int main(void) {
    RCC->AHB1ENR |= (1U << 0);
    GPIOA->MODER &= ~(3U << 10);
    GPIOA->MODER |=  (1U << 10);
    while (1) {
        GPIOA->ODR |=  (1U << 5);
        delay_ms(500);
        GPIOA->ODR &= ~(1U << 5);
        delay_ms(500);
    }
}
```

With BSRR

```c
#include "stm32f4xx.h"
void delay_ms(uint32_t ms) {
    for (uint32_t i=0;i<ms*4000;i++) __NOP();
}
int main(void) {
    RCC->AHB1ENR |= (1U << 0);
    GPIOA->MODER &= ~(3U << 10);
    GPIOA->MODER |=  (1U << 10);
    while (1) {
        GPIOA->BSRR = (1U << 5);
        delay_ms(500);
        GPIOA->BSRR = (1U << 21);
        delay_ms(500);
    }
}
```

Same effect, but the second version is atomic and one instruction shorter.

$5 \times 2 = 10$, so PA5's mode bits sit at position 10. $5 + 16 = 21$ is its reset bit in BSRR.

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The code below compiles and runs. PA5 stays dark. The registers read back as zero.

```
GPIOA->MODER &= ~(3U << 10);
GPIOA->MODER |=  (1U << 10);
GPIOA->ODR   ^=  (1U << 5);
```

Your prediction

- Is the C wrong?
- What would you check first?
- How can the debugger prove it?

The code below compiles and runs. PA5 stays dark. The registers read back as zero.

```
GPIOA->MODER &= ~(3U << 10);
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GPIOA->ODR   ^=  (1U << 5);
```

Your prediction

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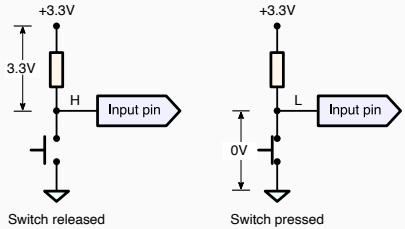
The answer

The C is fine. The clock is off: `RCC->AHB1ENR |= (1U « 0);` is missing. A peripheral with no clock ignores writes and reads back zero. Write a pattern to `MODER` in the debugger and read it back.

Experiment 3: Button Controls LED

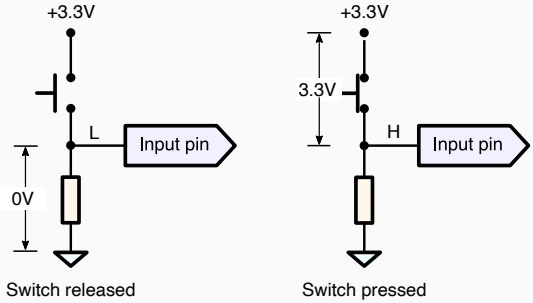
```c
#include "stm32f4xx.h"
int main(void) {
    RCC->AHB1ENR |= (1U << 2) | (1U << 0);  // GPIOC, GPIOA
    GPIOA->MODER &= ~(3U << 10);
    GPIOA->MODER |=  (1U << 10);        // PA5 output (LED)
    GPIOC->MODER &= ~(3U << (13*2));    // PC13 input
    GPIOC->PUPDR &= ~(3U << (13*2));
    GPIOC->PUPDR |=  (1U << (13*2));    // pull-up

    while (1) {
        if (!(GPIOC->IDR & (1U << 13))) {
            GPIOA->BSRR = (1U << 5);         // pressed -> on
        } else {
            GPIOA->BSRR = (1U << (5+16));    // released -> off
        }
    }
}
```



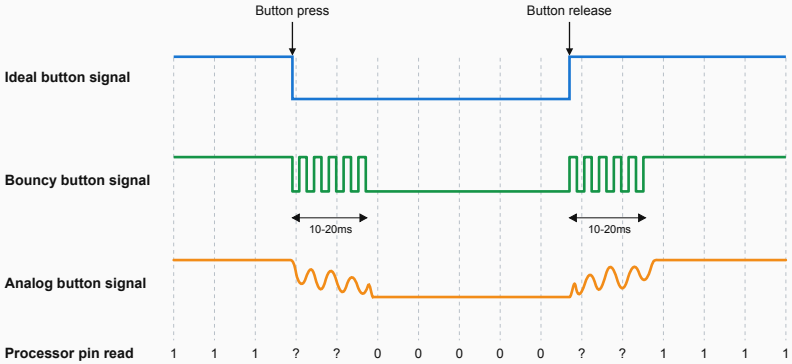
The button is active low: pressed reads `0`. When released, a pull-up holds the pin HIGH (the Nucleo has one on the board; the code adds the ~40 kΩ internal one). When pressed, the switch gives a low-resistance path to ground. That path wins.

The Other Convention: Pull-Down



A pull-down with an active-high switch is the mirror image: released reads LOW, and pressed connects the pin to V_{DD}. Use `PUPDR = 10` instead of `01`, and change the polarity in the code to match.

Experiment 4: Why a Button Needs Debouncing



A mechanical contact does not close cleanly. It bounces for several milliseconds. A processor that polls at MHz speed sees one press as dozens.

Debouncing

```c
#include "stm32f4xx.h"
void delay_ms(uint32_t ms) { for (uint32_t i=0;i<ms*4000;i++)
      __NOP(); }
void handle_press(void) { /* app-specific action */ }

int main(void) {
    RCC->AHB1ENR |= (1U << 2);         // GPIOC clock
    GPIOC->MODER &= ~(3U << (13*2));  // PC13 input
    GPIOC->PUPDR &= ~(3U << (13*2));
    GPIOC->PUPDR |=  (1U << (13*2));  // pull-up

    while (1) {
        if ((GPIOC->IDR & (1U << 13)) == 0) {
            delay_ms(20);              // let it settle
            if ((GPIOC->IDR & (1U << 13)) == 0) {
                // a real press
                while ((GPIOC->IDR & (1U << 13)) == 0)
                    ;                  // wait for release
                handle_press();
            }
        }
    }
}
```

Mind the brackets

In C, `&` binds *looser* than `==`. So `IDR & (1«13) == 0` is parsed as `IDR & ((1«13) == 0)`, which is always zero.

Always put brackets around the mask.

A 20 ms delay is the usual choice. It is long enough for a mechanical switch to settle, and short enough that a user does not notice.

Experiment 5: Four-Bit Binary Counter

Four LEDs on PB0–PB3 display a counter. Directly connects this chapter to your digital-electronics course.

```c
#include "stm32f4xx.h"
void delay_ms(uint32_t ms) { for (uint32_t i=0; i<ms*4000; i++) __NOP(); }

int main(void) {
    RCC->AHB1ENR |= (1U << 1);        // enable GPIOB clock
    GPIOB->MODER &= ~(0xFFU);         // clear PB0..PB3 (two bits each)
    GPIOB->MODER |=  (0x55U);         // PB0..PB3 as outputs

    uint8_t count = 0;
    while (1) {
        GPIOB->ODR = (GPIOB->ODR & ~0xFU) | (count & 0xFU);
        count++;
        delay_ms(300);
    }
}
```

Why the mask-and-write

`GPIOB->ODR` controls all 16 pins. The expression `(ODR & ~0xF) | (count & 0xF)` clears the low nibble and writes the new value there, leaving the other 12 pins untouched.

Experiment 6: Measuring Reaction Time

```c
#include "stm32f4xx.h"
void delay_ms(uint32_t ms) { for (uint32_t i=0;i<ms*4000;i++) __NOP(); }
volatile uint32_t reaction_ms;      // inspect this in the debugger
int main(void) {
        RCC->AHB1ENR |= (1U << 0) | (1U << 2);  // GPIOA, GPIOC
        GPIOA->MODER &= ~(3U << 10);
        GPIOA->MODER |=  (1U << 10);            // PA5 output (LED)
        GPIOC->MODER &= ~(3U << (13*2));        // PC13 input (button)
        GPIOC->PUPDR &= ~(3U << (13*2));
        GPIOC->PUPDR |=  (1U << (13*2));        // pull-up
        while (1) {
                GPIOA->BSRR = (1U << (5+16));   // LED off
                delay_ms(2000);                 // wait before the cue
                GPIOA->BSRR = (1U << 5);        // LED on: react now

                reaction_ms = 0;
                while (GPIOC->IDR & (1U << 13)) {   // until the button goes LOW
                        delay_ms(1);
                        reaction_ms++;
                }
                GPIOA->BSRR = (1U << (5+16));   // acknowledge
                delay_ms(1000);
        }
}
```

This measurement is only as accurate as `delay_ms`, a calibrated software loop. Its timing shifts with compiler optimization and clock speed. A hardware counter would make it exact.

Experiment 6: Knight Rider LED Chaser

```c
#include "stm32f4xx.h"
void delay_ms(uint32_t ms) { for (uint32_t i=0;i<ms*4000;i++) __NOP(); }

int main(void) {
        RCC->AHB1ENR |= (1U << 1);      // enable GPIOB clock
        GPIOB->MODER &= ~(0xFFFFU);     // clear PB0..PB7
        GPIOB->MODER |=  (0x5555U);     // PB0..PB7 as outputs (01 each)

        int pos = 0, dir = 1;
        while (1) {
                GPIOB->BSRR = (0xFFU << 16);   // all eight LEDs off
                GPIOB->BSRR = (1U << pos);     // light only the current one

                if (pos == 7 || pos == 0) dir = -dir;  // bounce at the ends
                pos += dir;

                delay_ms(80);
        }
}
```

One lit bit, walking back and forth

Only one output pin is ever high at a time; everything else is cleared first with a single atomic BSRR write, then the new bit is set with a second write. `pos` does the work a hardware shift register would do in an all-analogue version of this circuit — flipping `dir` at each end turns a shift into a bounce.

GPIO Init: Inputs with Pull-Ups and Outputs

```c
void GPIO_Init(void)
{
    /* Enable GPIOA and GPIOC clocks */
    RCC->AHB1ENR |= (1U << 0) | (1U << 2);

    /* Configure PA1 and PA2 as inputs */
    GPIOA->MODER &= ~((3U << 2) | (3U << 4));

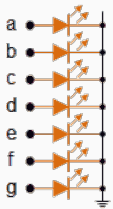
    /* Enable internal pull-up resistors for PA1 and PA2 */
    GPIOA->PUPDR &= ~((3U << 2) | (3U << 4));
    GPIOA->PUPDR |=  ((1U << 2) | (1U << 4));

    /* Configure PC0-PC7 as outputs */
    GPIOC->MODER &= ~0x0000FFFFU;
    GPIOC->MODER |=  0x00005555U;
}
```

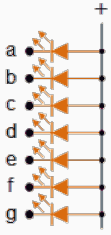
Two pins, one register, same pattern

PA1 and PA2 each need a clear-then-set on two `MODER` bits and two `PUPDR` bits — clearing first avoids OR-ing into whatever the reset value left behind. PC0–PC7 is the same idea done for eight pins at once: one 16-bit clear mask, one 16-bit set mask, instead of eight separate two-bit edits.

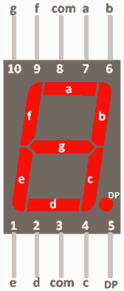
Experiment 7: A Seven-Segment Display



Common Cathode



Common Anode





Use a lookup table, not per-segment logic

Deciding which of seven segments to light for each digit is a table, not an algorithm. One array indexed by the digit replaces a whole page of `if` statements. It is also faster and easier to check.

Common-Anode Lookup Table

```c
#include "stm32f4xx.h"
void delay_ms(uint32_t ms) { for (uint32_t i=0;i<ms*4000;i++) __NOP()
      ; }

// common-anode codes, digits 0-9
// bit0=a ... bit6=g, bit7=dp -- 0 lights the segment
static const uint8_t seg7_ca[10] = {
    0xC0, 0xF9, 0xA4, 0xB0, 0x99,
    0x92, 0x82, 0xF8, 0x80, 0x90
};

int main(void) {
    RCC->AHB1ENR |= (1U << 1);   // enable GPIOB clock
    GPIOB->MODER &= ~(0xFFFFU);  // clear PB0..PB7
    GPIOB->MODER |=  (0x5555U);  // PB0..PB7 as outputs

    uint8_t digit = 0;
    while (1) {
        GPIOB->ODR = seg7_ca[digit];  // already active-low
        digit = (digit + 1) % 10;
        delay_ms(1000);
    }
}
```

Why the values look inverted

Common-anode means the shared pin is tied to VDD; each segment lights when its own pin is pulled LOW. So the table is the bitwise complement of the familiar common-*cathode* codes: `0` is `0x3F` for a cathode display but `0xC0` here.

Because the table is already active-low, driving it is a single `ODR` write — no inversion in the loop.

A Lookup Table for a 7-Segment Display

```
void Display_7Segment(uint8_t number)
{
    const uint8_t segment[10] =
    {
        0x3F,  /* 0 */
        0x06,  /* 1 */
        0x5B,  /* 2 */
        0x4F,  /* 3 */
        0x66,  /* 4 */
        0x6D,  /* 5 */
        0x7D,  /* 6 */
        0x07,  /* 7 */
        0x7F,  /* 8 */
        0x6F   /* 9 */
    };

    GPIOC->ODR = (GPIOC->ODR & ~0xFFU) | segment[number];
}
```

Trading logic for memory

Each byte in `segment[]` is a precomputed bit pattern for the seven segments, driven on the same PC0–PC7 bus just set up as outputs. The `& ~0xFF` / OR pattern touches only the low byte of `ODR`, so any other pins on port C stay untouched — the same read-modify-write discipline used everywhere else in this chapter.

Segment Mapping for `Display_7Segment`

- PC0 → a
- PC1 → b
- PC2 → c
- PC3 → d
- PC4 → e
- PC5 → f
- PC6 → g
- PC7 → decimal point (dp)

Why this matters

This mapping is what makes the `segment[]` table unambiguous: each bit position names one physical segment. The hex patterns assume active-HIGH outputs — correct for a common-cathode display, where a '1' sinks current through that segment's LED.

Driving a Pin, Viewed at Five Levels

Level	Representation
Physical hardware	PMOS/NMOS output pair charging the LED's series resistor
Register	`GPIOA_BSRR` bit 5, at `0x40020018`
C	`GPIOA->BSRR = (1U << 5);`
Assembly	`MOV R0,#0x20` / `STR R0,[R1,#0x18]`
Machine code	Two Thumb-2 encodings in Flash

Worth pausing over

This is the first time every level of the course model is filled in by something you built yourself. Read it as one chain, from the voltage on the pin up to the C statement.

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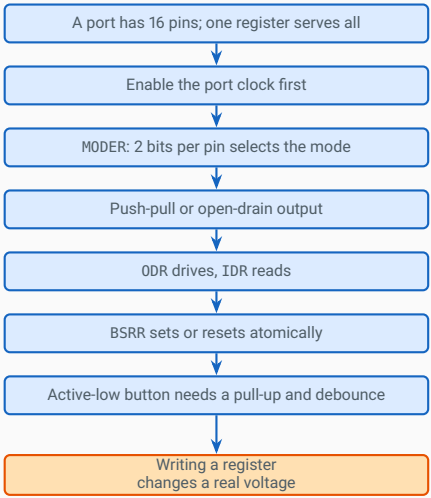
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Summary



Terminology covered

- Port, pin, `MODER`
- Push-pull, open-drain
- `ODR`, `IDR`, `BSRR`
- Pull-up, debouncing

Next: Chapter 9

Interrupts and Event-Driven Programming

Polling a button wastes the processor. Next, hardware tells us.

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